

A wide-angle photograph of a lush green rice field in Cambodia. The foreground is filled with tall, vibrant green rice stalks. In the middle ground, a small body of water reflects the warm, golden light of the sun, which is low on the horizon. Several palm trees and other tropical vegetation are scattered across the landscape. A faint grid pattern is overlaid on the entire image, suggesting a digital or technological theme.

Deep Learning-Based Rice Leaf Disease Classification

An End-to-End System for Agricultural Decision Support in Cambodia

Dara NEB

Faculty of Digital Technologies, American University of Phnom Penh (AUPP)

Rice is Cambodia's Lifeline, but Production is Vulnerable

- Rice is the primary staple crop, critical for national food security and rural livelihoods.
- Production is consistently threatened by diseases, leading to significant yield losses and reduced crop quality.
- Five major diseases are the primary culprits: Bacterial Leaf Blight, Leaf Blast, Brown Spot, Leaf Scald, and Sheath Blight.



The Core Challenge is an Expertise Gap

- Timely and accurate diagnosis is essential for effective disease management.
- However, many farmers lack access to plant pathology experts, leading to misdiagnosis and improper treatment.
- While AI solutions exist globally, there is a lack of localized, practical systems tailored for the specific context of Southeast Asian agriculture.



Limited Access
to Diagnosis

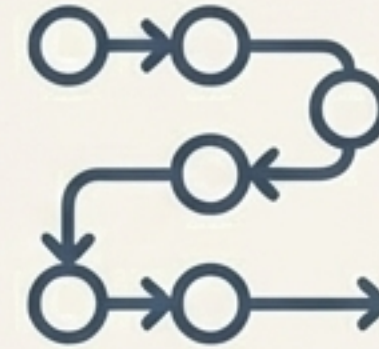


Project Objectives: Building a Deployable AI Pathologist



1. Develop an Accurate Model:

Build a deep learning model to classify the five major rice leaf diseases and healthy leaves from images.



2. Engineer a Full Pipeline:

Construct a complete machine learning pipeline, from data preparation to model evaluation.



3. Create an Accessible Interface:

Deploy the model in a simple, intuitive user interface using Streamlit for real-time classification.



4. Plan for Mobility: Save the final model in a format (.pth) that is ready for future export to offline mobile devices (Android/iOS).

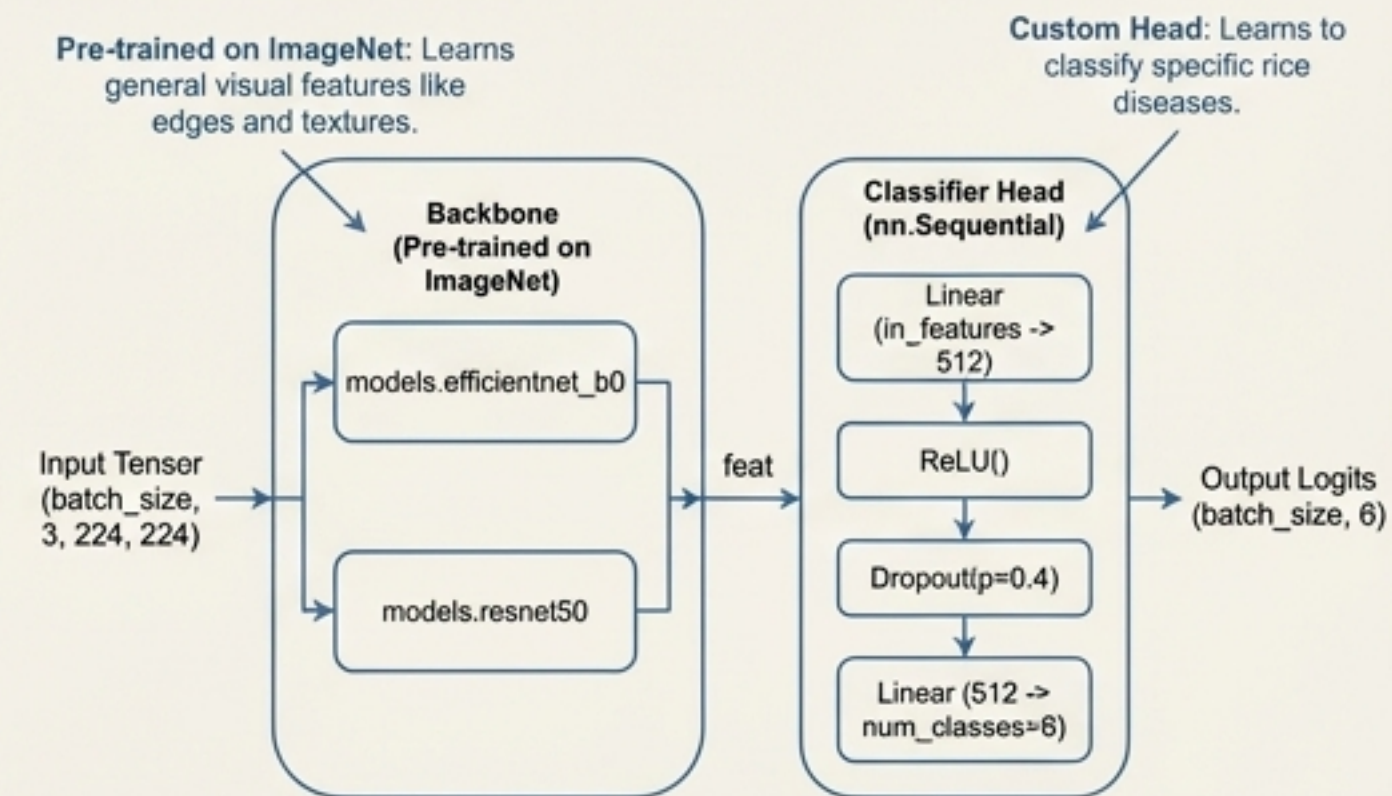
The Architectural Approach: A Baseline vs. a State-of-the-Art Model

THE DATASET

- **Source:** Public dataset of ~1,800 images.
- **Classes:** Bacterial Leaf Blight, Brown Spot, Leaf Blast, Leaf Scald, Sheath Blight, and Healthy Leaves.
- **Preprocessing:** All images resized to 224x224 pixels. Data augmentation (random rotations, flips, color jitter) was applied to enhance robustness and address class imbalances.

THE MODELS

- **1. RiceCNN (Baseline):** A custom-built, lightweight CNN with five convolutional layers, designed to establish a performance baseline baseline.
- **2. EfficientNet-B0 (Transfer Learning):** A high-performance model using a backbone pre-trained on ImageNet. Chosen for its superior trade-off between accuracy and computational efficiency.

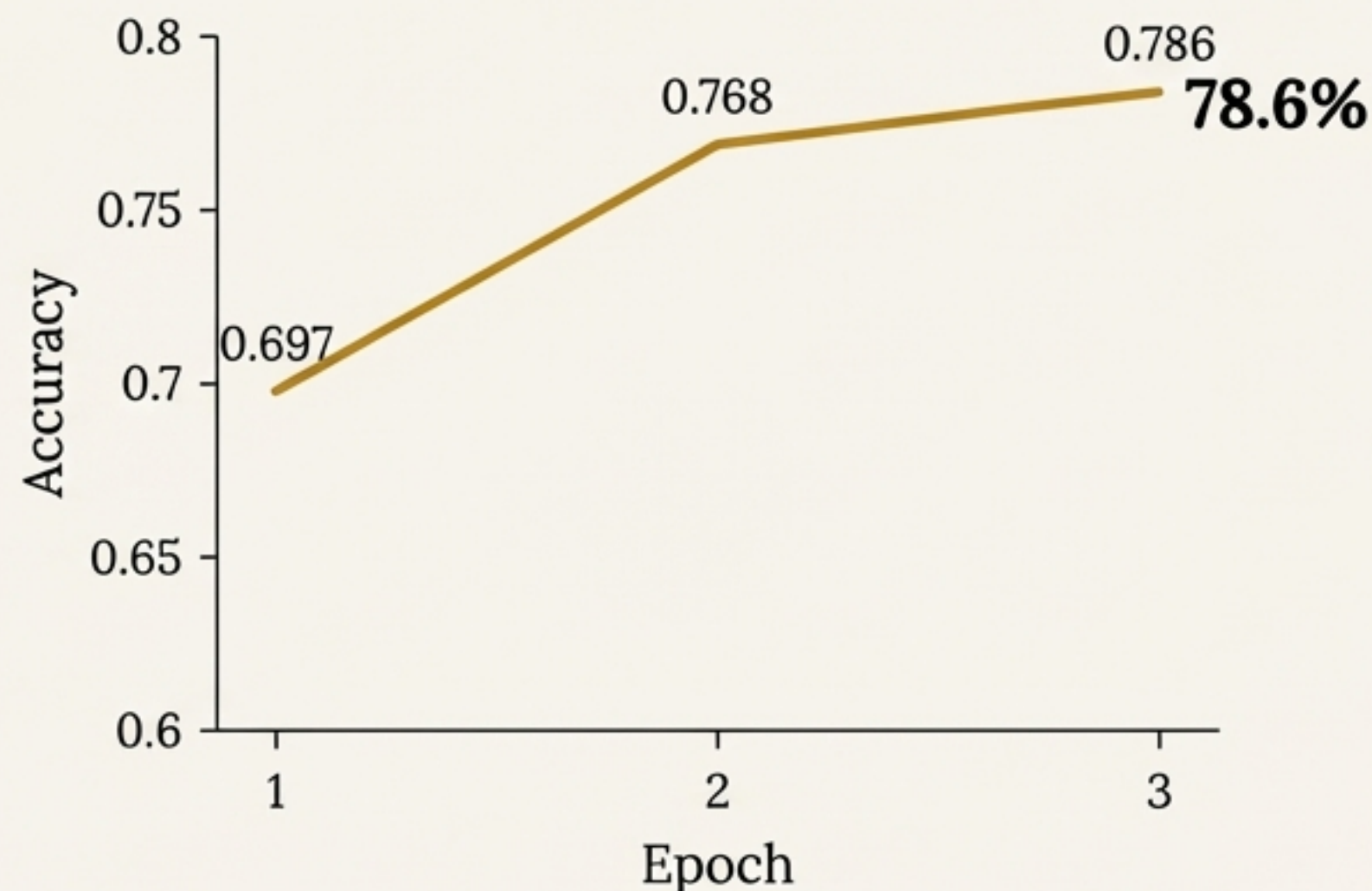


Stage 1: The Initial Hurdle

Training the classifier head with a frozen backbone reveals a performance plateau.

- In the first stage, only the final classification layers were trained. The pre-trained EfficientNet backbone was kept 'frozen'.
- This approach leverages general feature recognition learned from ImageNet but limits the model's ability to learn domain-specific patterns.

Validation Accuracy (Stage 1)



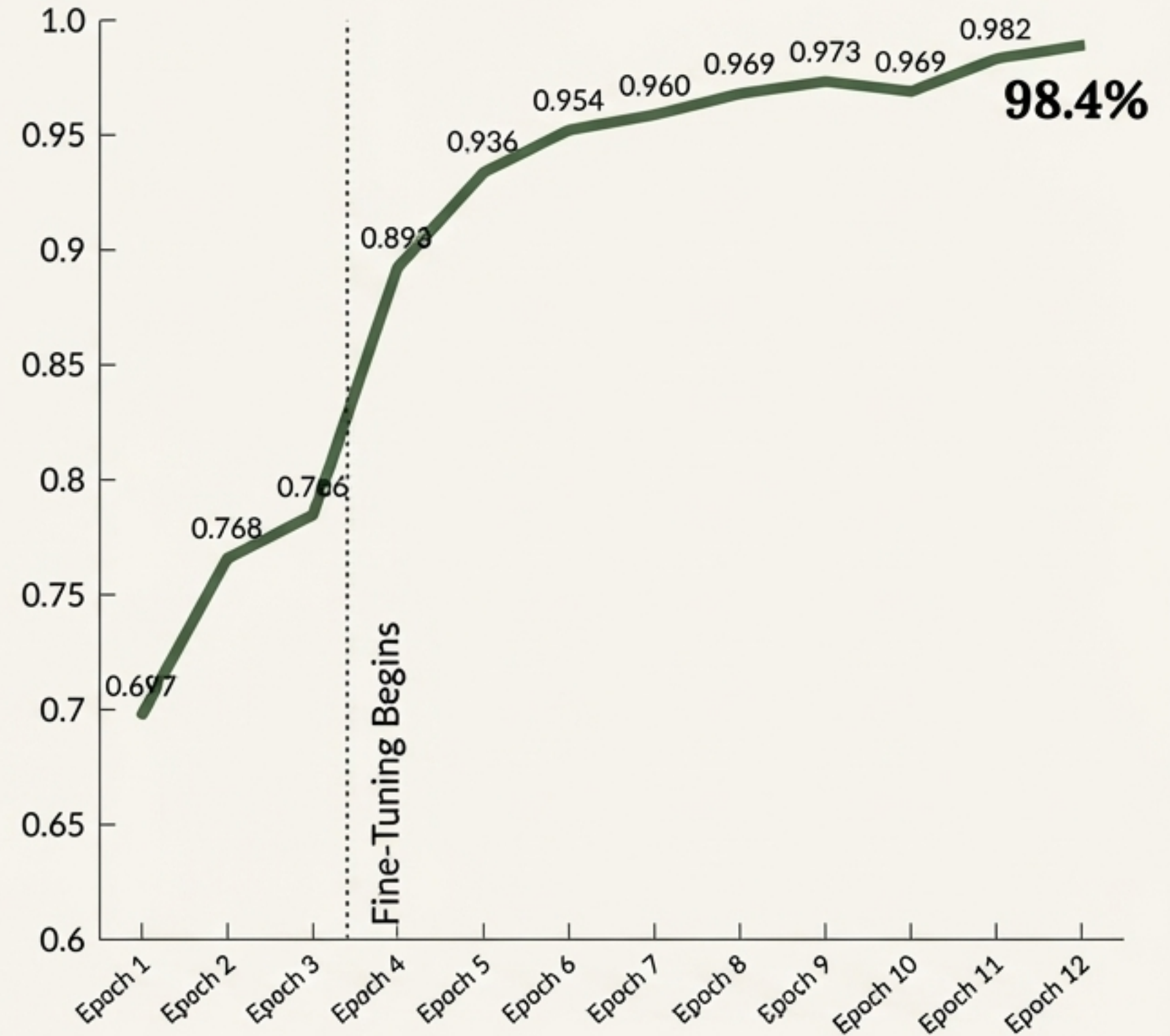
```
==== Stage 1: Training classifier only ====  
[Head] Epoch 1 Train Acc=0.558 Val Acc=0.697  
[Head] Epoch 2 Train Acc=0.673 Val Acc=0.768  
[Head] Epoch 3 Train Acc=0.721 Val Acc=0.786
```


The Breakthrough: Unlocking Potential with Fine-Tuning

- In Stage 2, the entire network, including the pre-trained backbone, was “unfrozen” and trained on the rice disease dataset.
- This allowed the model to adapt its deep-seated knowledge and learn the specific textures, color variations, and lesion patterns unique to rice diseases.

This is like taking a college student who already knows how to read (ImageNet knowledge) and handing them a new textbook on Rice Diseases. They learn the specific subject much faster and more deeply than a child starting from the alphabet (a custom CNN).

Validation Accuracy (Full Training)

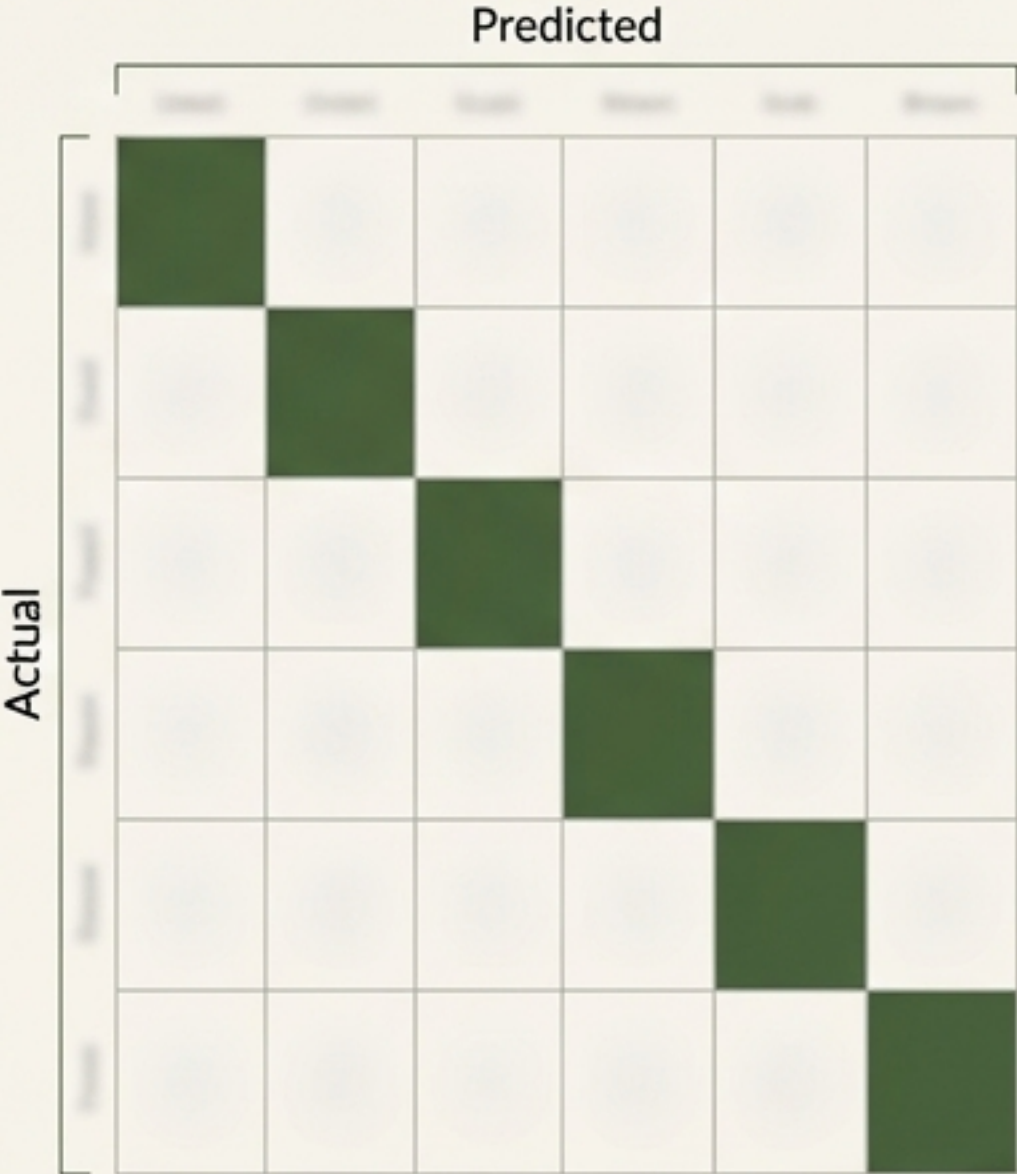


Final Performance Validates the Two-Stage Approach

Key Insight:

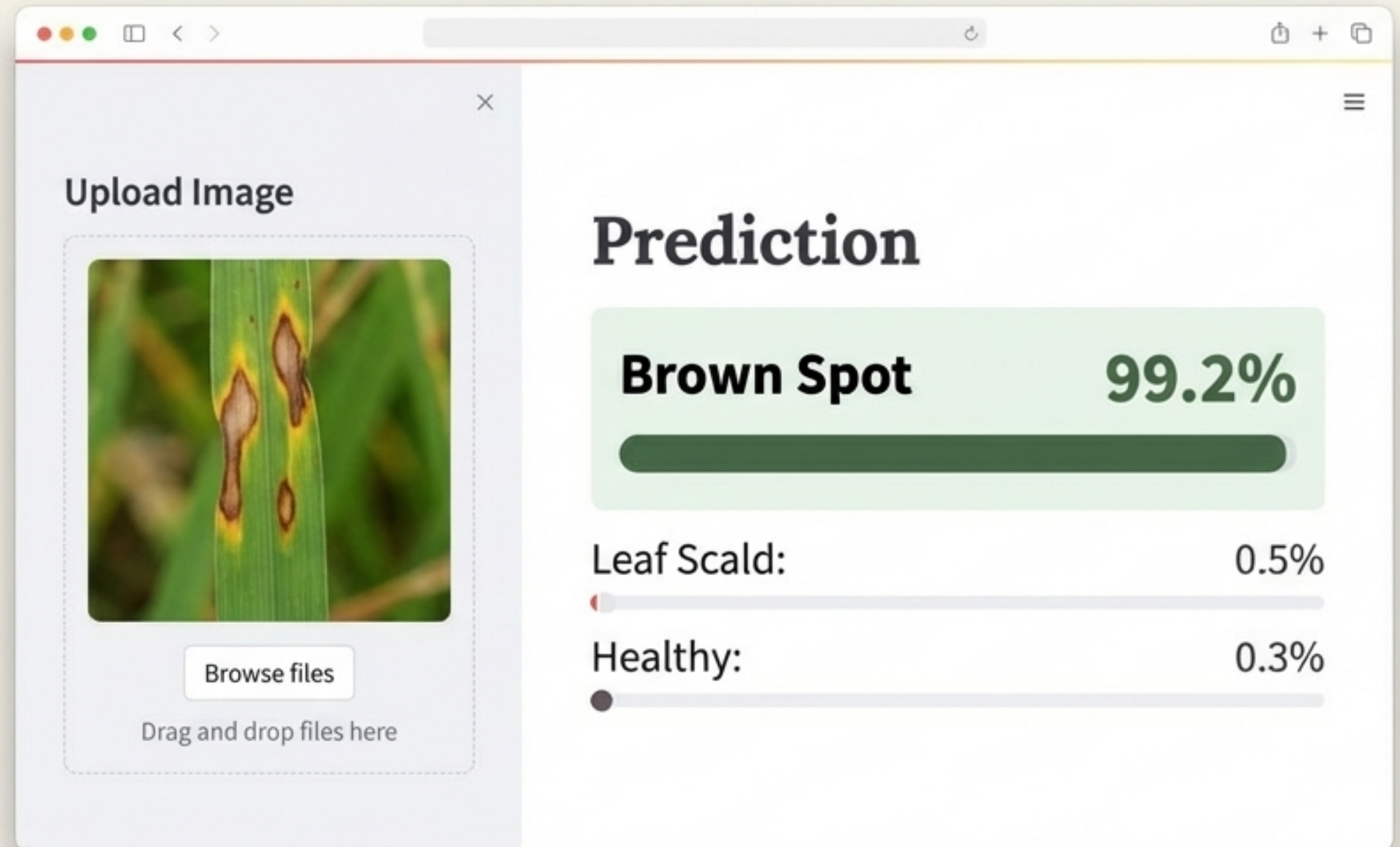
- The fine-tuning process was critical. It significantly reduced misclassifications between visually similar diseases (e.g., Leaf Scald vs. Brown Spot).
- This confirms that the model learned true, domain-specific features, improving feature separation and overall diagnostic power.

Metric	Result
Final Validation Accuracy	98.4%
Precision	High (per-class)
Recall	High (per-class)
F1-Score	High (per-class)



From Model to Tool: The RiceNet Streamlit Interface

The trained EfficientNet-B0 model was deployed into a simple and intuitive web-based graphical interface using Streamlit. This allows for camps icassification precees for rapid, real-time disease identification by uploading an image of a rice leaf.



The Next Chapter: Scaling Impact for Cambodian Farmers



Improve Generalization

Collect real-world field images from Cambodia to create a more robust, localized dataset.



Enhance Trust

Implement model explainability using Grad-CAM to visualize *why* the model makes a certain diagnosis.



Achieve True Accessibility

Develop a fully **offline mobile version** of the application, designed for use in rural areas with limited or no internet connectivity. This is the key next step to empower farmers directly in the field.